c-only slope law under the journal control law (J6)

($h := \bar{w} - \bar{w}_s$, $\bar{w}_s$ = visible surface, given ($\bar{w}_s = 0$ in every c-only run); $p = 2$ pinned; c = the one constant; $d = 2$; $e :=$ true $-$ estimate; all lengths $/R$, gain $/a_{nom}$, force $\times a_o$)

1 Law and control law

(gain law: unchanged from 0928 / 1006)

$$\bar{a}'(h;\, c) = \frac{1-c}{h^2}$$
$$\frac{\partial \bar{a}'}{\partial h} = -\frac{2\,\bar{a}'}{h}$$
$$\frac{\partial \bar{a}'}{\partial c} = -\frac{1}{h^2}$$
$$\frac{\partial \bar{a}'}{\partial \bar{a}_w} = 0$$

(control law; every estimate on the right is the posterior of step k, Section 4)

$$\bar{f}_d[k] = \frac{1}{\hat{\bar{a}}_w[k]}\Big(\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] - \hat{x}_D[k]\Big), \qquad \hat{x}_D \equiv 0 \quad \text{(J6)}$$

$$M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \qquad = \hat{\bar{a}}_w[k]\,\bar{f}_d[k]$$

$$\bar{f}_{\text{det}}[k] := \frac{\Delta\bar{w}_d[k]}{\hat{\bar{a}}_w[k]} \qquad \text{(the exogenous share of } \bar{f}_d\text{; regressor of } F_e)$$

(reference: the law this file replaces, `motion_control_law_formC_b.m:1773`; $\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$)

$$\bar{f}_d^{\text{old}}[k] = \frac{1}{\hat{\bar{a}}_w[k-1]}\Big(\Delta\bar{w}_d[k] + (1-\lambda_c)\Big[\delta\bar{w}_m[k] + \nabla_d\bar{w}_d[k] - \sum_{i=1}^{d}\hat{\bar{a}}_w[k-i]\,\bar{f}_d^{\text{old}}[k-i]\Big]\Big)$$

2 State

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ c\,\big]^\top \qquad \text{(code slots 1–5; 6, 7 locked, } 7 = p = 2\text{)} \quad \text{(J14)}$$

3 True dynamics

($\bar{a}'_{\text{true}}$ = the plant's slope; $w_T[k]$ = thermal step, $\operatorname{Var} w_T[k] = \kappa_T\,\bar{a}_w[k]$)

$$
\begin{aligned}
\bar{w}[k+1] &= \bar{w}[k] + \bar{a}_w[k]\,\bar{f}_d[k] + w_T[k] \quad \text{(J5)}\\
\delta\bar{w}_3[k] &= \bar{w}_d[k] - \bar{w}[k]\\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k]\\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k]\\
\delta\bar{w}_3[k+1] &= \delta\bar{w}_3[k] + \Delta\bar{w}_d[k] - \bar{a}_w[k]\,\bar{f}_d[k] - w_T[k]\\
&= \lambda_c\,\delta\bar{w}_3[k] + (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] - \bar{f}_d[k]\,e_{\bar{a}_w}[k] - w_T[k] \quad \text{(J7, 8, 14)}\\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}\big(\bar{w}[k] - \bar{w}_s\big)\,\Delta\bar{w}[k]\\
c[k+1] &= c[k]\\
\Delta\bar{w}[k] &:= \bar{w}[k+1] - \bar{w}[k] = M[k] + \bar{f}_d[k]\,e_{\bar{a}_w}[k] + w_T[k]
\end{aligned}
$$

(the two identities behind row 3)

$$
\begin{aligned}
\bar{a}_w[k]\,\bar{f}_d[k] &= \big(\hat{\bar{a}}_w[k] + e_{\bar{a}_w}[k]\big)\bar{f}_d[k] = M[k] + \bar{f}_d[k]\,e_{\bar{a}_w}[k]\\
\Delta\bar{w}_d[k] - M[k] &= -(1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] = -(1-\lambda_c)\big(\delta\bar{w}_3[k] - e_{\delta\bar{w}_3}[k]\big)
\end{aligned}
$$

4 Estimator

($\hat{h}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] - \bar{w}_s$; $\bar{a}'[k] = (1-\hat{c}[k])/\hat{h}[k]^2$; superscript $-$ = before the update with $y[k]$)

(a) time update $k-1 \to k$, input $M[k-1]$ known

$$
\begin{aligned}
\delta\hat{\bar{w}}_1^-[k] &= \delta\hat{\bar{w}}_2[k-1]\\
\delta\hat{\bar{w}}_2^-[k] &= \delta\hat{\bar{w}}_3[k-1]\\
\delta\hat{\bar{w}}_3^-[k] &= \lambda_c\,\delta\hat{\bar{w}}_3[k-1] \quad \text{(J16)}\\
\hat{\bar{a}}_w^-[k] &= \hat{\bar{a}}_w[k-1] + \big(1-\hat{c}[k-1]\big)\Big(\frac{1}{\hat{h}[k-1]} - \frac{1}{\hat{h}[k-1]+M[k-1]}\Big)\\
&\qquad (= \hat{\bar{a}}_w[k-1] + \bar{a}'[k-1]\,M[k-1] + O(M^2))\\
\hat{c}^-[k] &= \hat{c}[k-1]
\end{aligned}
$$

(b) measurement update with $y[k]$

$$\hat{\mathbf{x}}[k] = \hat{\mathbf{x}}^-[k] + L[k]\big(\mathbf{y}[k] - \hat{\mathbf{y}}^-[k]\big) \quad \text{(J16, 19)}$$

(c) force of step k, from the posterior of (b)

$$\bar{f}_d[k] = \frac{M[k]}{\hat{\bar{a}}_w[k]}, \qquad M[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \quad \text{(J6)}$$

5 Error dynamics

($h - \hat{h} = -e_{\delta\bar{w}_3}$; $\Delta\bar{w} - M = \bar{f}_d\,e_{\bar{a}_w} + w_T$; r' = family residual of the slope on this wall)

$$\bar{a}'_{\text{true}} - \bar{a}'[k] = \frac{2\,\bar{a}'[k]}{\hat{h}[k]}\, e_{\delta\bar{w}_3}[k] - \frac{1}{\hat{h}[k]^2}\, e_c[k] + r'[k]$$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= e_{\delta\bar{w}_3}[k] - \bar{f}_d[k]\, e_{\bar{a}_w}[k] - w_T[k] \quad \text{(J17, 18)} \\
e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] + \big(\bar{a}'_{\text{true}} - \bar{a}'[k]\big)\Delta\bar{w}[k] + \bar{a}'[k]\big(\Delta\bar{w}[k] - M[k]\big) \\
&= \frac{2\,\bar{a}'[k]}{\hat{h}[k]} M[k]\, e_{\delta\bar{w}_3}[k] + \big(1 + \bar{a}'[k]\,\bar{f}_d[k]\big) e_{\bar{a}_w}[k] - \frac{M[k]}{\hat{h}[k]^2}\, e_c[k] + \bar{a}'[k]\, w_T[k] + M[k]\, r'[k] \\
e_c[k+1] &= e_c[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & -\bar{f}_{\text{det}}[k] & 0 \\
0 & 0 & \dfrac{2\,\bar{a}'[k]}{\hat{h}[k]} M[k] & 1 + \bar{a}'[k]\,\bar{f}_{\text{det}}[k] & -\dfrac{M[k]}{\hat{h}[k]^2} \\
0 & 0 & 0 & 0 & 1
\end{bmatrix} \quad \text{(J18)}$$

($\bar{f}_{\text{det}}$ in place of $\bar{f}_d$ in F_e: the share $(1-\lambda_c)\,\delta\hat{\bar{w}}_3/\hat{\bar{a}}_w$ of $\bar{f}_d$ is a function of the estimate and is correlated with $e_{\delta\bar{w}_3}$; the same convention as $F_{dw,\text{det}}$ in the file this replaces)

(what changes against the law this file replaces, $F_{dw}[k] := \bar{f}_d[k] + (1-\lambda_c)\big(\bar{f}_d[k-1] + \bar{f}_d[k-2]\big)$)

$$\begin{aligned}
F_{e,33}: \quad \lambda_c &\;\to\; 1 \\
F_{e,34}: \quad -F_{dw} &\;\to\; -\bar{f}_{\text{det}} \\
F_{e,43}: \quad \bar{a}'(1-\lambda_c) + \frac{2\,\bar{a}'}{\hat{h}} M &\;\to\; \frac{2\,\bar{a}'}{\hat{h}} M \\
F_{e,44}: \quad 1 + \bar{a}' F_{dw} &\;\to\; 1 + \bar{a}'\bar{f}_{\text{det}}
\end{aligned}$$

6 Measurement

($\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$; y_1 = the delayed tracking-error reading; y_2 = the gain reading of Section 7)

$$\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \qquad = \delta\bar{w}_m[k] \quad \text{(J15)} \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k] \quad \text{(J13, 15)} \\
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \bar{a}'[k]\,\nabla_d \bar{w}_d[k] \qquad \text{(the walk-back of the journal's two-step delay, (J21) note)}
\end{aligned}$$

$$H[k] = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 \\
0 & 0 & -\dfrac{2\,\bar{a}'[k]}{\hat{h}[k]} \nabla_d \bar{w}_d[k] & 1 & \dfrac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^2}
\end{bmatrix} \quad \text{(J15)}$$

(as implemented: y_2 whitened with $s = a_{cov}$, echo factor $1 - S[k]$; unchanged from 1006 Section 7)

$$y_2[k] = \bar{a}_m[k] - (1 - a_{cov})\,\bar{a}_m[k-1]$$
$$\hat{y}_2[k] = s\Big(\hat{\bar{a}}_w[k] - (1 - S[k])\,\bar{a}'[k]\,\nabla_d \bar{w}_d[k]\Big)$$
$$H_2[k] = s\,(1 - S[k])\Big[\,0,\ 0,\ -\frac{2\,\bar{a}'[k]}{\hat{h}[k]}\nabla_d \bar{w}_d[k],\ 1,\ \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^2}\Big], \qquad \frac{\partial \hat{y}_2}{\partial \bar{a}_w} - H_{24} = s\,S[k]$$

7 Gain reading from the loop variance

(the journal's (J9)–(J13) chain, with (J11)(J12) re-derived for THIS loop; position block only, $\bar{a}_w$ frozen, steady state; $q := \kappa_T\,\bar{a}_w$, $r := \sigma^2_{n_w}$)

(J9)(J10), as in code `:1746--1748`

$$\overline{\delta \bar{w}}_m[k+1] = a_{pd}\,\delta \bar{w}_m[k] + (1 - a_{pd})\,\overline{\delta \bar{w}}_m[k] \quad \text{(J9)}$$
$$\delta \bar{w}_r[k] = \delta \bar{w}_m[k] - \overline{\delta \bar{w}}_m[k+1]$$
$$\hat{\sigma}^2_{\delta \bar{w}_r}[k+1] = a_{cov}\,\delta \bar{w}_r[k]^2 + (1 - a_{cov})\,\hat{\sigma}^2_{\delta \bar{w}_r}[k] \quad \text{(J10)}$$

steady-state filter of the position block ($F_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, $H_3 = [\,1\ 0\ 0\,]$, $Q_{33} = q$, $R_1 = r$)

$$m := P^-_{11} \qquad m = \frac{m\,r}{m + r} + q \quad \Longrightarrow \quad m^2 - q\,m - q\,r = 0, \qquad m = \frac{q + \sqrt{q^2 + 4\,q\,r}}{2}$$
$$K_1 := \frac{m}{m + r} \qquad = L_{11} = L_{21} = L_{31}$$
$$p_1 := P_{11} = \frac{m\,r}{m + r} = K_1\,r$$
$$q = m - p_1 = \frac{m^2}{m + r}$$
$$P^- = \begin{bmatrix} m & m & m \\ m & m + q & m + q \\ m & m + q & m + 2q \end{bmatrix}, \qquad P = \begin{bmatrix} p_1 & p_1 & p_1 \\ p_1 & p_1 + q & p_1 + q \\ p_1 & p_1 + q & p_1 + 2q \end{bmatrix}$$

tracking error of the loop ($e_{\bar{a}_w} = 0$; $\nu[k] := y_1[k] - \hat{y}^-_1[k]$, white, $\mathrm{Var}\,\nu = m + r$; $\mathrm{E}[\delta \hat{\bar{w}}_3\,e_{\delta \bar{w}_3}] = 0$)

$$\delta \bar{w}_3[k+1] = \lambda_c\,\delta \hat{\bar{w}}_3[k] + e_{\delta \bar{w}_3}[k] - w_T[k]$$
$$\delta \hat{\bar{w}}_3[k+1] = \lambda_c\,\delta \hat{\bar{w}}_3[k] + K_1\,\nu[k+1]$$
$$\mathrm{Var}\,\delta \hat{\bar{w}}_3 = \frac{K_1^2\,(m + r)}{1 - \lambda_c^2} = \frac{q}{1 - \lambda_c^2}$$
$$\mathrm{Var}\,\delta \bar{w}_3 = \mathrm{Var}\,\delta \hat{\bar{w}}_3 + P_{33} = \Big(2 + \frac{1}{1 - \lambda_c^2}\Big)q + K_1\,r \quad \text{(J11)}$$
$$\mathrm{Var}\,\delta \bar{w}_m = \mathrm{Var}\,\delta \bar{w}_3 + r = \Big(2 + \frac{1}{1 - \lambda_c^2}\Big)q + (1 + K_1)\,r \quad \text{(J12)}$$

(J13) for this loop; K_1 evaluated at the estimate

$$C_{dpmr} = 2 + \frac{1}{1-\lambda_c^2} \qquad \text{(unchanged)}$$

$$C_n[k] = 1 + K_1\big(\kappa_T\,\hat{\bar{a}}_w[k],\ \sigma_{n_w}^2\big) \qquad \Big(\text{replaces } \frac{2}{1+\lambda_c}\Big)$$

$$\bar{a}_m[k] = \frac{\hat{\sigma}^2_{\delta\bar{w}_r}[k] - C_n[k]\,\sigma_{n_w}^2}{C_{dpmr}\,\kappa_T} \quad \text{(J13)}$$

(composition with (J9)(J10) and the colour of $\hat{\sigma}^2_{\delta\bar{w}_r}$: from the full autocovariance of the loop, $\mathbf{z} = [\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \delta\hat{\bar{w}}_1,\ \delta\hat{\bar{w}}_2,\ \delta\hat{\bar{w}}_3\,]^\top$, gain ratio $g := \bar{a}_w/\hat{\bar{a}}_w$ in the loop)

$$\mathbf{z}[k+1] = A(\lambda_c, K_1, g)\,\mathbf{z}[k] + B\begin{bmatrix} w_T[k] \\ n_w[k+1] \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -g(1-\lambda_c) \\ 0 & K_1 & 0 & 0 & 1-K_1 & 0 \\ 0 & K_1 & 0 & 0 & -K_1 & 1 \\ 0 & K_1 & 0 & 0 & -K_1 & \lambda_c \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & K_1 \\ 0 & K_1 \\ 0 & K_1 \end{bmatrix}$$

$$\Sigma = A\,\Sigma\,A^\top + B\,\mathrm{diag}(q,\ r)\,B^\top, \qquad R_\delta(\tau) = \big[A^\tau\Sigma\big]_{11}, \qquad \mathrm{Var}\,\delta\bar{w}_m = \Sigma_{11} + r$$

$C_{dpmr,\mathrm{eff}}(K_1),\ C_{n,\mathrm{eff}}(K_1),\ IF(K_1)$: (J9)(J10) applied to $R_\delta(\tau)$

$$S_T(K_1),\ S_n(K_1) = \frac{\partial \ln \mathrm{Var}\,\delta\bar{w}_m}{\partial \ln g}\Big|_{g=1} \quad \text{thermal / noise share (echo)}$$

8 Noise

$$Q[k] = \kappa_T\,\hat{\bar{a}}_w[k]\,\mathbf{g}\mathbf{g}^\top, \qquad \mathbf{g} = \big[\,0,\ 0,\ -1,\ \bar{a}'[k],\ 0\,\big]^\top, \qquad Q_{55} = 0 \quad \text{(J17, 21)}$$

$$R[k] = \mathrm{diag}\big(\sigma_{n_w}^2,\ R_2[k]\big) \quad \text{(J19)}$$

$$\mathrm{E}\big[w_T[k]\,n_w[j]\big] = 0, \qquad \mathrm{E}\big[w_T[k]\,w_T[j]\big] = 0\ (k \neq j)$$

(what changes against the law this file replaces)

$$Q_{33}^{\mathrm{old}} = \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\sum_{i=1}^{d}\hat{\bar{a}}_w[k-i]\Big) + (1-\lambda_c)^2\,\sigma_{n_w}^2 \ \rightarrow\ Q_{33} = \kappa_T\,\hat{\bar{a}}_w[k]$$

MA(2) memory (m_1, m_2) $\rightarrow$ none

correlated (w_T, n_w) $\rightarrow$ none

(KF recursion, unchanged)

$$L[k] = P^-[k]\,H^\top[k]\big(H[k]\,P^-[k]\,H^\top[k] + R[k]\big)^{-1} \quad \text{(J19)}$$

$$P[k] = \big(I - L[k]\,H[k]\big)P^-[k] \quad \text{(J20)}$$

$$P^-[k+1] = F_e[k]\,P[k]\,F_e^\top[k] + Q[k] \quad \text{(J21)}$$

9 Observability (gates 1–3 on paper)

Gate 1, the two columns of c (unchanged):

$$F_{e,45}[k] = -\frac{M[k]}{\hat{h}[k]^2}, \qquad H_{25}[k] = \frac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^2}$$

Gate 2, time signatures:

$$e_c[k+1] = e_c[k]$$

$$e_{\bar{a}_w}[k+1] = \big(1 + \bar{a}'\bar{f}_{\text{det}}\big)e_{\bar{a}_w} - \frac{M}{\hat{h}^2}\, e_c + \ldots \qquad \text{(level, drifting with the height travelled)}$$

$$e_{\delta\bar{w}_3}[k+1] = e_{\delta\bar{w}_3} - \bar{f}_{\text{det}}\, e_{\bar{a}_w} - w_T, \qquad y_1[k+2] = e_{\delta\bar{w}_3}[k] + \ldots$$

(a random walk read by y_1 after d steps; was λ_c-contracting)

Gate 3, $\det O_2$ with coefficients frozen ($\alpha := 1 + \bar{a}'\bar{f}_{\text{det}}$, $\beta := M/\hat{h}^2$, $\gamma := \nabla_d \bar{w}_d/\hat{h}^2$):

$$\text{pair } (e_{\bar{a}_w}, e_c),\ \text{row } y_2: \quad \det\begin{bmatrix} 1 & \gamma \\ \alpha & \gamma - \beta \end{bmatrix} = -\beta - \gamma(\alpha - 1) = -\frac{M + \nabla_d \bar{w}_d\, \bar{a}'\bar{f}_{\text{det}}}{\hat{h}^2}$$

$$\text{pair } (e_{\bar{a}_w}, e_c),\ \text{row } y_1: \quad \det\begin{bmatrix} -\bar{f}_{\text{det}} & 0 \\ -\bar{f}_{\text{det}}\,\alpha & \bar{f}_{\text{det}}\,\beta \end{bmatrix} = -\bar{f}_{\text{det}}^2\,\beta = -\frac{\bar{f}_{\text{det}}^2\, M}{\hat{h}^2}$$

(in a hold $\nabla_d \bar{w}_d = 0$ and $\bar{f}_{\text{det}} = 0$: the y_1 row vanishes, the y_2 row is $-M/\hat{h}^2$ with $M = (1-\lambda_c)\,\delta\hat{\bar{w}}_3$, noise-driven and zero-mean; both $\to 0$ as $\hat{h} \to \infty$. M in $F_{e,43}$, $F_{e,45}$ keeps its $(1-\lambda_c)\,\delta\hat{\bar{w}}_3$ share, as in the file this replaces; only $F_{e,34}$, $F_{e,44}$ use $\bar{f}_{\text{det}}$)

Seed and prior

(unchanged from the f1p2 arm except the position block)

$$\hat{c}[0] = c_{\text{plane}} = +0.0533 \qquad \text{(plane init, wall-test protocol)}$$

$$\hat{\bar{a}}_w[0] = D_\perp\Big(\frac{1}{\bar{w}[0] - \bar{w}_s}\Big) \qquad \text{(plane polynomial at the start height)}$$

$$\sigma_{c,0} = 0.6 \qquad \text{(wall family: plane } 0.05 \ldots \text{cell } 0.64)$$

$$P_{44}[0] = \Big(\frac{\partial \bar{a}}{\partial c}\Big)^2 \sigma_{c,0}^2 + P_{a,\text{floor}}, \qquad P_{45}[0] = \frac{\partial \bar{a}}{\partial c}\,\sigma_{c,0}^2, \qquad P_{a,\text{floor}} = 0.002^2$$

$$P_{55}[0] = \sigma_{c,0}^2, \qquad Q_{55} = 0$$

position block (closed form of Section 7 at $q = \kappa_T\,\hat{\bar{a}}_w[0]$, $r = \sigma_{n_w}^2$; was a DARE with $F_{3,33} = \lambda_c$):

$$P_{1:3,1:3}[0] = \begin{bmatrix} p_1 & p_1 & p_1 \\ p_1 & p_1 + q & p_1 + q \\ p_1 & p_1 + q & p_1 + 2q \end{bmatrix}, \qquad p_1 = K_1\, r$$